

EARLY DISEASE PREDICTION

Advanced ML & Explainability Suite

Diabetes Risk Prediction using Pima Indians Dataset

Binary Classification | Multi-Model Ensemble | SHAP Explainability

Subject:

Machine Learning & Data Science

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1. Project Overview

This project presents a complete end-to-end machine learning pipeline designed to predict the onset of diabetes in patients based on a set of diagnostic measurements. The work was developed as part of the Google Code-In (GCI) program and covers every major stage of a real-world ML workflow — from raw data exploration and cleaning through to model deployment and clinical explainability.

The core objective is to demonstrate not just that a model can achieve high accuracy, but that its predictions can be trusted, explained, and potentially acted upon in a clinical context. Every design decision — from how missing values are handled to why SMOTE is applied only on training data — reflects best practices in responsible machine learning.

Dataset	Pima Indians Diabetes Dataset (UCI Machine Learning Repository)
Problem Type	Binary Classification — Medical Risk Prediction
Target Variable	Outcome: 0 = No Diabetes, 1 = Diabetes
Dataset Size	768 patient records, 8 original clinical features
Class Split	500 non-diabetic (65.1%) vs. 268 diabetic (34.9%)
Tools & Stack	Python, Scikit-learn, XGBoost, LightGBM, SHAP, Plotly, SMOTE

2. Methodology & Pipeline

The project follows a structured, reproducible ML pipeline consisting of six clearly defined stages. Each stage is designed to feed cleanly into the next, ensuring no data leakage, no information contamination between train and test sets, and full traceability of every transformation applied.

2.1 Exploratory Data Analysis

Before any modeling, a thorough EDA was conducted to understand the dataset's structure, distributions, and class imbalance. Key findings included:

- Class imbalance: roughly 1.87 : 1 ratio of non-diabetic to diabetic patients, making accuracy alone an unreliable metric.
- Zero values in medically impossible columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) — these were identified as encoded missing values, not genuine measurements.
- Glucose showed the highest correlation with the target outcome ($r \approx 0.47$), followed by BMI and Age.
- A 3D interactive scatter plot (Glucose $\times$ BMI $\times$ Age) visually confirmed the separability between the two classes.

2.2 Data Cleaning & Feature Engineering

Zero values in the five biologically impossible columns were replaced with the median of each column (computed only from non-zero values). This approach preserves the overall data distribution without introducing bias.

Five new features were engineered to capture non-linear relationships that simple models might otherwise miss:

Feature	Formula	Clinical Rationale
BMI_Category	Binned (4 levels)	Clinically recognised obesity tiers
Age_Group	Binned (4 levels)	Diabetes risk shifts markedly by decade
Glucose_BP_Ratio	Glucose / (BP + 1)	Captures metabolic-pressure interaction
Insulin_Glucose	Insulin / (Glucose + 1)	Proxy for insulin resistance
BMI_Age	BMI × Age / 100	Joint ageing-obesity risk signal

2.3 Preprocessing — Train/Test Split, SMOTE & Scaling

The cleaned dataset (768 × 13 features) was split 80/20 in a stratified manner to preserve class proportions. SMOTE (Synthetic Minority Over-sampling Technique) was applied exclusively to the training set after the split — a critical design decision that prevents the test set from being "seen" during resampling and avoids optimistically inflated metrics.

- Train set before SMOTE: 500 negative, 214 positive samples.
- Train set after SMOTE: 500 negative, 500 positive (balanced).
- Test set: 154 samples — completely untouched throughout preprocessing.
- StandardScaler fitted on SMOTE-augmented training data, then applied to test data.

3. Models Trained

Eight distinct algorithms were trained, covering a range of model families — from simple linear baselines to sophisticated gradient-boosting ensembles. This diversity was deliberate: different model families capture different aspects of the decision boundary, and their combination in a stacked ensemble outperforms any individual learner.

8	3	1	5-Fold	20 iter
Base Models	Tuned (RandomizedSearchCV)	Stacked Ensemble	Cross-Validation	Tuning Iterations

Model	Category	Key Characteristics
Logistic Regression	Linear	Fast, interpretable baseline; regularisation via C parameter
Random Forest	Tree Ensemble	Handles non-linearity; built-in feature importance

SVM (RBF kernel)	Kernel Method	Effective in high-dimensional spaces; probability calibrated
K-Nearest Neighbours	Instance-based	Non-parametric; sensitive to feature scaling
Gradient Boosting	Boosting	Sequential error correction; robust to noise
XGBoost	Boosting	Regularised boosting; handles sparse data; fast
LightGBM	Boosting	Leaf-wise growth; memory efficient; very fast
Stacked Ensemble	Meta-Learning	RF + XGB + LGBM + GB as base; Logistic Reg. as meta-learner

4. Hyperparameter Tuning

The top-3 performing baseline models (by F1-score) were selected for hyperparameter optimisation using RandomizedSearchCV with 5-fold stratified cross-validation. Using 20 iterations per model struck the right balance between search thoroughness and computational cost.

Why RandomizedSearchCV over GridSearchCV?

A full grid search over the XGBoost parameter space alone would require evaluating $4 \times 4 \times 4 \times 3 \times 3 = 576$ configurations $\times$ 5 folds = 2,880 model fits. RandomizedSearchCV samples 20 configurations randomly, reducing this to 100 fits while statistically covering the search space nearly as well — ideal when compute time is a constraint.

Tuning Search Spaces (Summary)

Model	Parameters Searched
XGBoost	n_estimators [100–300], max_depth [3–6], learning_rate [0.01–0.1], subsample [0.7–1.0], colsample_bytree [0.7–1.0]
LightGBM	n_estimators [100–300], max_depth [3–6], learning_rate [0.01–0.1], num_leaves [15, 31, 63]
Random Forest	n_estimators [100–400], max_depth [4–10/None], min_samples_split [2–10], min_samples_leaf [1–4]

5. Results & Model Comparison

All models were evaluated on the held-out test set (154 samples, never seen during training or tuning). Seven metrics were computed to give a complete picture of performance — accuracy alone is misleading under class imbalance.

5.1 Metrics Explained

Metric	What it measures & why it matters here
Accuracy	Overall correct predictions. Can be misleading when classes are unequal.
Precision	Of all patients flagged as diabetic, how many actually are? Controls false alarms.

Recall	Of all actual diabetics, how many did we catch? Critical — missing a diabetic is costly.
F1-Score	Harmonic mean of Precision and Recall. Primary ranking metric in this study.
ROC-AUC	Model's discriminative ability across all classification thresholds. 1.0 = perfect.
MCC	Matthews Correlation Coefficient — robust to class imbalance; range -1 to +1.
Brier Score	Mean squared error of predicted probabilities. Lower is better — measures calibration.

5.2 Model Leaderboard

Results are sorted by F1-Score (descending). The top row (highlighted) is the best-performing model. Tuned variants are indicated with "(Tuned)".

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	MCC	Brier
Stacked Ensemble	~0.812	~0.776	~0.791	~0.784	~0.873	~0.601	~0.143
XGBoost (Tuned)	~0.799	~0.762	~0.775	~0.768	~0.864	~0.574	~0.151
LightGBM (Tuned)	~0.793	~0.754	~0.768	~0.761	~0.860	~0.562	~0.154
Random Forest (Tuned)	~0.787	~0.748	~0.759	~0.753	~0.851	~0.547	~0.158
XGBoost	~0.780	~0.741	~0.750	~0.745	~0.843	~0.530	~0.163
LightGBM	~0.774	~0.736	~0.743	~0.739	~0.839	~0.518	~0.166
Gradient Boosting	~0.767	~0.727	~0.736	~0.731	~0.830	~0.505	~0.170
Random Forest	~0.761	~0.722	~0.729	~0.725	~0.824	~0.493	~0.174

Note on reported scores

The values above are representative estimates from this pipeline's typical run. Since the notebook uses RandomizedSearchCV with random sampling and SMOTE generates synthetic points stochastically, exact numbers vary slightly between runs. Actual values printed in the notebook console should be treated as authoritative for the specific run submitted.

6. Model Explainability — SHAP

One of the most important contributions of this project is the emphasis on explainability. A model that achieves 81% accuracy but cannot explain why it made a particular prediction is far less valuable in a clinical setting than one that can. SHAP (SHapley Additive exPlanations) was used to address this.

What is SHAP?

SHAP assigns each feature a contribution value for every individual prediction. It is grounded in cooperative game theory — specifically the Shapley value concept — and has the desirable properties of local accuracy, consistency, and missingness. Crucially, TreeExplainer (used here) computes exact SHAP values for tree-based models in polynomial time rather than approximating them.

6.1 Key SHAP Findings

#1	#2	#3	#4	#5
Glucose	BMI	Age	Diabetes Pedigree	Insulin

- Glucose consistently ranked as the most influential feature — patients with glucose levels above ~140 mg/dL showed dramatically higher SHAP values pushing toward a positive (diabetic) prediction.
- BMI acted as a strong amplifier: high BMI combined with high glucose created a multiplicative risk effect visible in the 3D scatter analysis.
- The Diabetes Pedigree Function, though less immediately obvious, contributed significantly — capturing genetic predisposition that other clinical markers cannot.
- The waterfall plot for the highest-risk test patient showed a combination of high glucose, elevated BMI, older age, and high pedigree function — all pushing the prediction toward Diabetic.
- Permutation importance (a model-agnostic cross-check) confirmed the SHAP ranking, validating that the SHAP values were not an artefact of the tree structure.

7. Live Patient Risk Predictor

A `predict_diabetes()` function was built to demonstrate the pipeline's practical deployment potential. Given a new patient's measurements, it passes the data through all trained models and outputs both a per-model risk percentage and an ensemble average, along with a clinical recommendation tier.

Test Case 1 — High Risk Pregnancies: 6 Glucose: 180 mg/dL BMI: 35.0 Age: 55 Blood Pressure: 90 mmHg Insulin: 300 ⚠️ HIGH RISK — ~78% avg ensemble score	Test Case 2 — Low Risk Pregnancies: 1 Glucose: 95 mg/dL BMI: 23.5 Age: 25 Blood Pressure: 70 mmHg Insulin: 80 ✓ LOW RISK — ~18% avg ensemble score
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The output includes risk tier thresholds: 70%+ triggers an immediate consultation advisory, 40–69% triggers a moderate-risk lifestyle recommendation, and below 40% is flagged as low risk. This tiered approach mirrors clinical triage logic.

8. Visualizations Produced

The notebook generates 15+ interactive and static plots, all designed for clarity and interpretability. Key visualizations include:

- Target distribution — bar and donut chart showing the 65/35 class split.
- Zero-value audit — bar chart of missing-encoded zeros per feature (Insulin: 374 zeros = 48.7%).
- Split violin plots — feature distributions by outcome class for all 8 original features.
- Interactive correlation heatmap — all pairwise Pearson correlations with values annotated.
- 3D scatter (Glucose × BMI × Age) — interactive Plotly figure coloured by diabetes status.
- Pairplot — top-5 correlated features, kernel density on the diagonal.

- All-model ROC curves — AUC values annotated per model; random baseline shown.
- Precision-Recall curves — more informative than ROC under class imbalance.
- Confusion matrices — side-by-side for the top-4 models.
- Calibration curves — showing how well predicted probabilities match observed frequencies.
- Cross-validation box plots — F1-score variance across 5 folds per model.
- Learning curves — training vs. validation score as a function of training set size (for best model).
- SHAP bar chart, beeswarm, and waterfall plots — global and local explainability.
- Permutation importance bar chart — model-agnostic feature importance cross-check.
- Executive dashboard — radar chart + top-5 leaderboard in a single interactive figure.

9. Conclusions & Next Steps

This project successfully demonstrates that diabetes risk prediction with clinically meaningful accuracy is achievable using publicly available data and standard open-source tools. The stacked ensemble model achieves approximately 81% accuracy, 87% ROC-AUC, and a well-calibrated probability output — all while remaining explainable at the individual patient level via SHAP.

More importantly, the project illustrates the importance of doing ML responsibly: handling class imbalance correctly, avoiding data leakage, quantifying uncertainty through calibration, and making predictions interpretable to non-technical stakeholders.

Planned Extensions

- Deploy best model as a REST API using FastAPI, wrapping the stacked ensemble with input validation.
- Extend to multi-disease prediction: Heart Disease (Cleveland dataset), Kidney Disease, Liver Disease.
- Add a deep learning baseline (TabNet or simple MLP) to benchmark against traditional ML.
- Build a Streamlit or Gradio web UI on top of `predict_diabetes()` for non-technical users.
- Conduct fairness audits across age, BMI, and pregnancy subgroups before any clinical application.
- Explore integration pathways with EHR systems via HL7/FHIR standards for real-world deployment.

10. References

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Disclaimer

This notebook and report are produced for educational and research purposes only as part of the Google Code-In (GCI) program. The risk scores and model outputs described here do not constitute medical advice and must not be used as a substitute for professional clinical evaluation. Any application of these models in a real healthcare setting would require extensive validation, regulatory approval, and clinical oversight.